Histology of the Pancreas

LG 3 · Dr Hero K. Mustafa
GIT BLOCK

Histology GIT Block · MEDucation by Mattin Majid

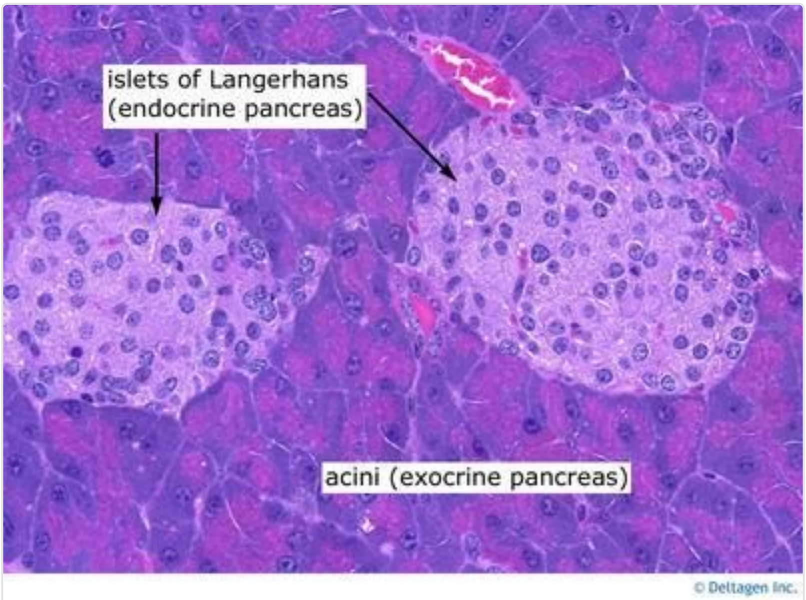
BEFORE YOU READ — ANSWER FROM MEMORY

1. What are the 2 main cell types in the exocrine acinus, and what does each one make?
2. Put the 4 pancreatic duct segments in order, from the acinus to the duodenum.
3. Name the 4 islet cell types and what each one secretes.

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

1 Pancreas — overview

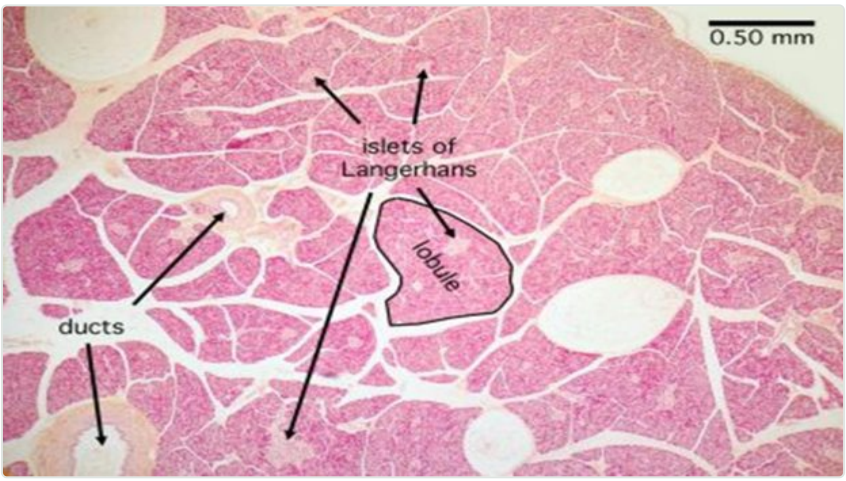
A **mixed exocrine & endocrine** gland. Enzymes are stored/released by the exocrine portion (cells + ducts); hormones are synthesised in the **islets of Langerhans**.



Islets of Langerhans (pale, endocrine) sitting among acini (exocrine).

2 Exocrine part — architecture

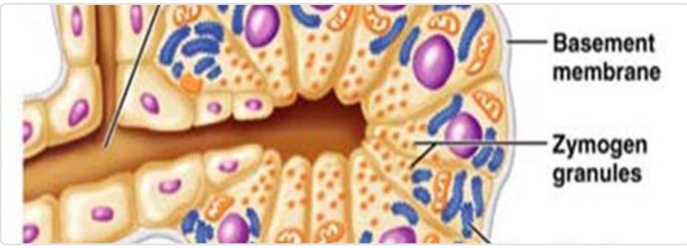
Capsule & septa	Thin loose c.t. capsule; septa extend inward, dividing the gland into lobules & lobes, carrying blood vessels and nerves
Gland type	Compound acinar gland
Function	Secretes enzymes central to digestion of carbohydrates, proteins & fats

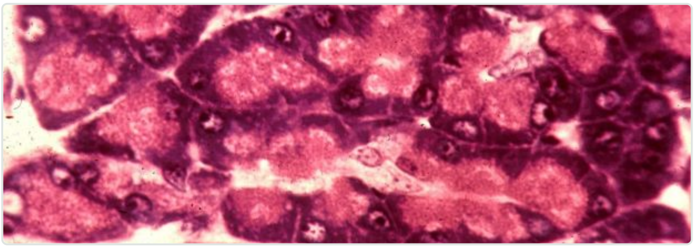


Lobule outlined, with ducts and islets of Langerhans visible between acini.

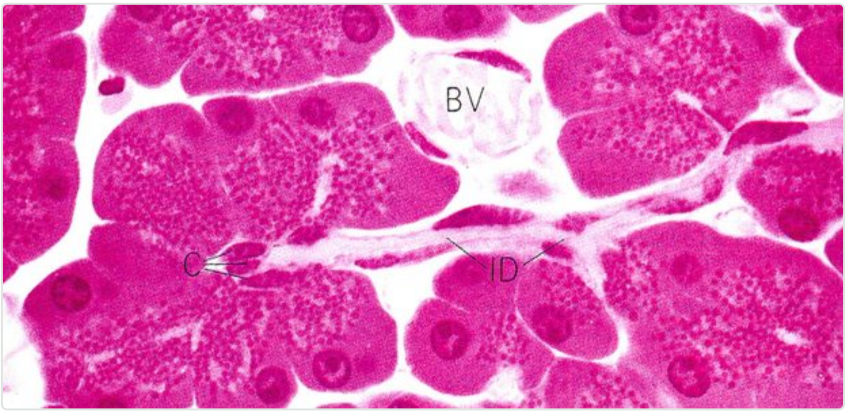
3 Exocrine cell types — ID table

Cell type	Location	Key ID feature	Secretes / controlled by
Acinar cell	Forms the acinus — several pyramidal cells per acinus	Basophilic **base** (RER + ribosomes); eosinophilic **apical** zymogen granules (enzymes packaged in membrane-bound granules)	Amylase (carbohydrates); lipase, phospholipase A2, cholesterol esterase (fats); proteases — trypsin, chymotrypsin, carboxypeptidase, elastase (proteins). Controlled by **cholecystokinin (CCK)** from duodenal enteroendocrine cells
Centroacinar cell	1–2 per acinus, inside the acinus **lumen**, at the origin of the intercalated duct	Condensed nucleus, clear cytoplasm	Watery, bicarbonate-rich fluid — neutralises acidic chyme, optimises enzyme activity. Controlled by **secretin** from small-intestinal enteroendocrine cells





Left: acinar cell diagram — basal RER, apical zymogen granules. Right: acinar cells on H&E.

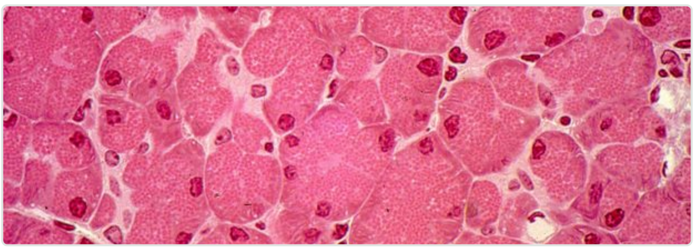
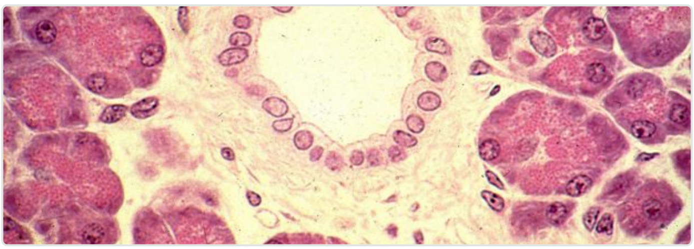


Centroacinar cells (pale, central) sitting inside the acinus lumen, at the start of the intercalated duct.

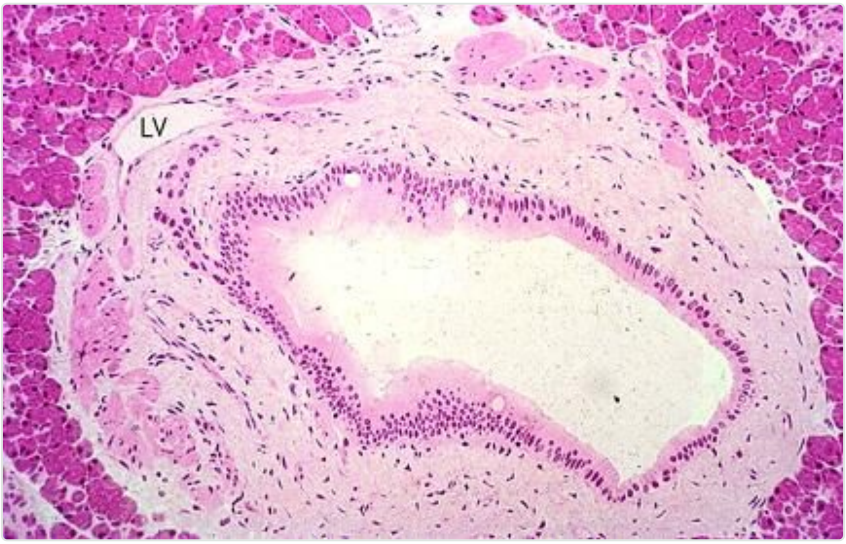
4 Duct system — the exam-favourite sequence

Acinar cells → intercalated duct → intralobular duct → interlobular duct → main pancreatic duct → duodenum.

Segment	Formed by	Epithelium
1. Intercalated duct	Smallest duct; begins directly from centroacinar cells inside the acinus	Simple squamous or low cuboidal
2. Intralobular duct	Multiple intercalated ducts joining within one lobule	Low columnar
3. Interlobular duct	Found between lobules, in the septa; surrounded by abundant c.t.	Columnar; **may contain goblet cells**
4. Main pancreatic duct	Convergence of many interlobular ducts; runs tail → head; joins the common bile duct	—; opens at the major duodenal papilla (**ampulla of Vater**), gated by the **sphincter of Oddi**

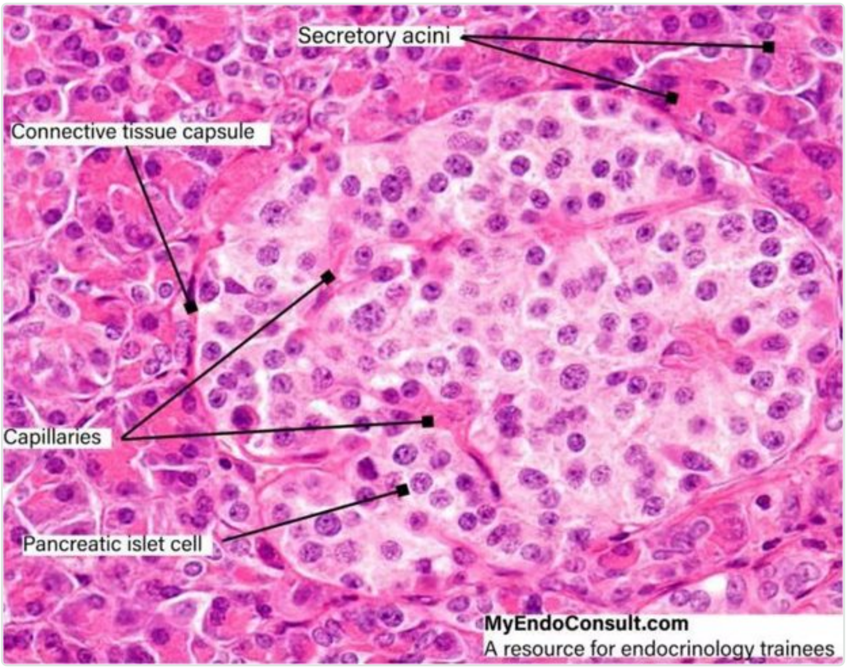
Left: intercalated duct. Right: intralobular duct, with connective tissue and exocrine (acinar) cells around it.



Interlobular duct — columnar epithelium with a goblet cell, set in abundant connective tissue.

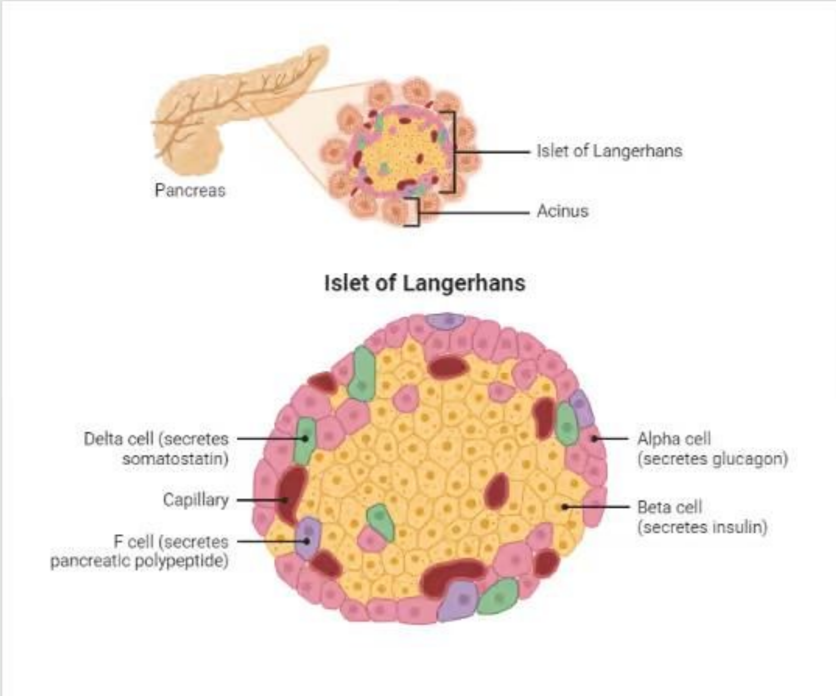
5 Endocrine portion — Islets of Langerhans

Appearance	Clusters of rounded cells, most numerous in the **tail** of the pancreas
Vs. exocrine cells	Smaller, paler, polygonal; cytoplasm has secretory granules containing hormones
Vasculature	Cells separated by a network of **fenestrated capillaries**
Capsule	Each islet is surrounded by reticular c.t. fibres



An islet (pancreatic islet cells, capillaries, connective tissue capsule) among surrounding secretory acini.

6 Islet cell types — the core discriminator



The four islet cell types and their position within the islet.

Cell type	Position	Secretes	% of islet	Function
A (alpha)	Periphery	Glucagon	~20%	Increases blood glucose
B (beta)	Centre	Insulin	~70%	Decreases blood glucose
D (delta)	Variable	Somatostatin	—	Inhibits both glucagon & insulin secretion
F / PP cell	Variable	Pancreatic polypeptide	—	Inhibits pancreatic exocrine secretion (enzymes & bicarbonate); relaxes gallbladder & decreases bile secretion

P Must-remember pearls

- Acinar cells = basophilic **base** (RER) + eosinophilic **apical** zymogen granules; secrete amylase, lipases and proteases under **CCK** control.
- Centroacinar cells sit inside the acinus lumen at the start of the intercalated duct; they make bicarbonate-rich fluid under **secretin** control.
- Duct order: acinus → **intercalated** duct → **intralobular** duct → **interlobular** duct (may have goblet cells) → **main pancreatic duct** → ampulla of Vater.
- **Beta cells** (insulin, ~70%, centre) and **alpha cells** (glucagon, ~20%, periphery) are the two dominant islet populations.
- Islets of Langerhans are most numerous in the **tail** of the pancreas.

Q Test yourself

1. A pancreatic cell with a basophilic base rich in RER and eosinophilic apical zymogen granules, secreting digestive enzymes under CCK control, is the:

A) Centroacinar cell
B) Acinar cell
C) Beta cell
D) Ductal cell

2. Which duct segment lies within the lobule, formed by several intercalated ducts joining, and lined by low columnar epithelium?

A) Interlobular duct
B) Main pancreatic duct
C) Intralobular duct
D) Intercalated duct

3. A watery, bicarbonate-rich secretion that neutralises acidic chyme is produced by:

A) Acinar cells, under CCK control
B) Centroacinar cells, under secretin control
C) Beta cells, under glucose control
D) Alpha cells, under glucagon control

4. Which islet cell type sits centrally, makes up ~70% of the islet, and secretes insulin?

A) Alpha cell
B) Beta cell
C) Delta cell
D) F (PP) cell

5. Somatostatin, which inhibits both insulin and glucagon secretion, is produced by:

A) Alpha cells
B) Beta cells
C) Delta cells
D) F (PP) cells

ANSWERS — COVER UNTIL YOU'VE COMMITTED

1. B — Acinar cell — basal RER + apical eosinophilic zymogen granules; CCK-controlled.

2. C — Intralobular duct — formed within a lobule by joining intercalated ducts; low columnar.

3. B — Centroacinar cell, secretin — watery bicarbonate-rich fluid neutralises chyme.

4. B — Beta cell — central, ~70% of islet, secretes insulin.

5. C — Delta cell — somatostatin inhibits both glucagon and insulin.

SPACED REVIEW

☐ Day 1	☐ Day 3	☐ Day 7	☐ Day 21	☐ Pre-exam